

AQUA-LENS

ABSTRACT

AquaLens is an IoT-based visual monitoring system developed using an AI Thinker ESP32-CAM. The system captures images of the water environment at regular intervals and stores them on a MicroSD card for monitoring and future analysis. The ESP32-CAM is also connected to Wi-Fi, providing a foundation for remote monitoring.

The system follows a simple workflow of camera initialization, MicroSD initialization, Wi-Fi connection, image capture, image storage, and repetition every 30 seconds. This provides an automated and low-cost method for collecting visual information from water environments. The captured images can be further used for monitoring visible changes and can be extended in the future with AI-based analysis and real-time alerts.

System Workflow:



Input:

- Water environment in front of the camera.
- Images captured using the ESP32-CAM.
- Wi-Fi network credentials.
- MicroSD card for image storage.

Working Principle:

- ESP32-CAM is powered ON.
- Camera is initialized and configured.
- MicroSD card is detected and initialized.
- ESP32-CAM connects to the configured Wi-Fi network.
- Camera captures an image.
- Captured image is stored as **photo.jpg** on the MicroSD card.
- System waits for **30 seconds**.
- The image capture process repeats automatically.

Output:

- Captured JPEG image stored on the MicroSD card.
- Wi-Fi connection status displayed on the Serial Monitor.
- ESP32-CAM IP address displayed.
- Image capture and storage status displayed.
- A sequence of visual observations can be collected over time.